

1 Introduction

This project analyzes over **3.7 million Citi Bike trips** collected between December 2024 and April 2025. Each trip represents a directed edge from a start station to an end station, allowing the Citi Bike system to be modeled as a weighted, directed network.

The purpose of our analysis is to understand:

- the overall structure and shape of NYC’s mobility network,
- the most influential stations and busiest corridors, and
- the community structure of the city as revealed by mobility patterns.

These findings support real-world operational planning, station siting, rebalancing, resilience, and prioritization of infrastructure investments.

2 Dataset Description

We combined four months of Citi Bike trip data, producing:

- **3,718,069 total trips**
- **2,298 unique stations** after cleaning
- **555,022 unique origin–destination pairs**

Each row corresponds to a directed trip:

start station $\rightarrow$ end station, with weight = Number of trips.

We cleaned the data by:

- treating station IDs as strings (to avoid float parsing issues),
- dropping rows with missing station IDs,
- removing self-loops (trips that start and end at the same station).

This produced a reliable network suitable for analysis.

3 Outcome A: Structural Properties of the Citi Bike Network

3.1 Global Network Statistics

The combined four-month network exhibits the following characteristics:

- **Nodes:** 2,298 stations
- **Edges:** 555,022 unique origin–destination pairs
- **Density:** 0.105 (relatively sparse overall)
- **Average in/out degree:** 241.52
- **Average in/out strength:** 1617.96
- **Reciprocity:** 0.633
- **Strongly connected components (SCC):** 71
- **Largest SCC size:** 2,228 stations
- **Weakly connected components (WCC):** 2
- **Largest WCC size:** 2,296 stations
- **Average shortest path:** 2.11
- **Diameter:** 5
- **Transitivity:** 0.518
- **Average clustering:** 0.5008

3.2 Interpretation

The Citi Bike network forms a **small-world mobility structure**: despite its size, the average shortest path is only about 2.1, and the clustering coefficient is high. This means:

- The system is **highly navigable**.
- Stations naturally form **local clusters** (e.g., neighborhoods).
- Most stations are only a few hops apart, ensuring smooth connectivity.

The high reciprocity (0.63) reflects common commuting patterns where riders travel in both directions along similar corridors.

3.3 Degree and Strength Distributions

In-degree, out-degree, in-strength, and out-strength distributions showed **heavy-tailed behavior**. A small number of stations handle a disproportionate share of total trips, confirming the presence of major activity hubs.

3.4 Monthly Structural Stability

Monthly statistics remained consistent across December through April. Although winter months had reduced station activity, the **overall structure, clustering, and path lengths remained stable**. This suggests that NYC's bicycle mobility patterns are resilient seasonally.

4 Outcome B: Hubs, Corridors, and Influence

Outcome B identifies stations that serve as the most active, most influential, and most structurally critical locations in the Citi Bike network.

4.1 Strength-Based Hubs

Strength (in + out trips) identifies stations with the heaviest overall use. The top hubs included:

- **W 21 St & 6 Ave**
- **University Pl & E 14 St**
- **West St & Chambers St**
- **11 Ave & W 41 St**
- **Lafayette St & E 8 St**

These stations form the **core operational backbone**. They require the most attention for:

- rebalancing,
- dock capacity,
- expansion planning,
- congestion mitigation.

4.2 PageRank Hubs

Weighted PageRank identified stations that are **important because they connect to other important stations**. Top PageRank hubs included:

- 11 Ave & W 41 St
- West St & Chambers St
- Melrose Ave & E 150 St
- Central Park S & 6 Ave

These stations are structurally significant beyond just volume.

4.3 Betweenness Centrality

Betweenness was computed on a backbone of the top 100,000 heaviest edges to focus on major corridors. The highest-betweenness stations included:

- W 21 St & 6 Ave
- Queens Plaza North & Crescent St
- Vernon Blvd & 36 Ave
- Franklin St & Dupont St

These stations act as **bridges between major regions**, especially across boroughs. Their failure would fragment travel.

4.4 Top Origin–Destination Corridors

The strongest OD corridors revealed NYC’s **bike superhighways**, including:

- Midtown east–west links,
- Lower Manhattan commuter flows,
- Williamsburg ↔ Manhattan,
- LIC ↔ Manhattan connections.

These corridors highlight where protected lanes should be expanded.

4.5 Weekday vs Weekend Dynamics

- Weekdays: dominated by commuter-heavy hubs in Midtown, Union Square, LIC.
- Weekends: recreational zones such as Central Park and waterfront areas rise in ranking.

This distinction is essential for planning rebalancing strategies.

5 Outcome C: Community Structure of NYC’s Mobility Network

5.1 Major Communities

Community detection using the Louvain method (or greedy modularity when unavailable) revealed:

- **6 major communities**
- Sizes: 821, 667, 423, 382, 3, and 2 stations
- **Modularity:** 0.523 (strong community structure)

5.2 Spatial Interpretation

Communities corresponded closely to NYC boroughs:

- **Community 0 (821 stations):** Brooklyn (Williamsburg, Bushwick, Bed-Stuy)
- **Community 1 (667 stations):** Manhattan core (Midtown, Chelsea, SoHo)
- **Community 2 (423 stations):** Upper Manhattan and South Bronx
- **Community 3 (382 stations):** Queens (LIC, Astoria, Sunnyside)

Tiny communities (size 2–3) represented isolated or peripheral stations.

5.3 Cross-Community Bridge Stations

Stations with highest cross-community flow included:

- **Queens Plaza North & Crescent St**
- **Vernon Blvd & 36 Ave**
- **Franklin St & Dupont St**

These act as **critical connectors across boroughs** and represent infrastructure chokepoints.

5.4 Monthly Community Stability

The number of communities and their sizes remained consistent across months, indicating that mobility neighborhoods are stable even as seasonal usage fluctuates.

6 Real-World Impact and Conclusions

The combined results across Outcomes A, B, and C reveal:

- The Citi Bike system is a **highly connected, small-world network**.
- A small number of stations form the **core operational backbone**.
- Community structure mirrors real borough boundaries and travel patterns.
- Several stations act as **fragile bridges** whose disruption would fragment mobility.
- The strongest OD corridors highlight where NYC should expand bike lanes and station capacity.

6.1 How This Helps the City

- **Operations:** better rebalancing and inventory management.
- **Planning:** identify where to expand station capacity.
- **Infrastructure:** target protected bike lanes on core corridors.
- **Resilience:** fortify high-betweenness stations and bridges.
- **Equity:** understand which neighborhoods are underserved.